deep se
dependable evolvable pervasive software engineering group

Exercise Session – Calendar

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Copy Control and Resource Management

- Classes that manage resources that do not reside in the class must define the copy-control members
- In order to define these members, we first have to decide what copying an object of our type will mean. In general, we have two choices:
  - **like-a-value:** the class behaves like a value
  - **like-a-pointer**: the class behaves like a pointer

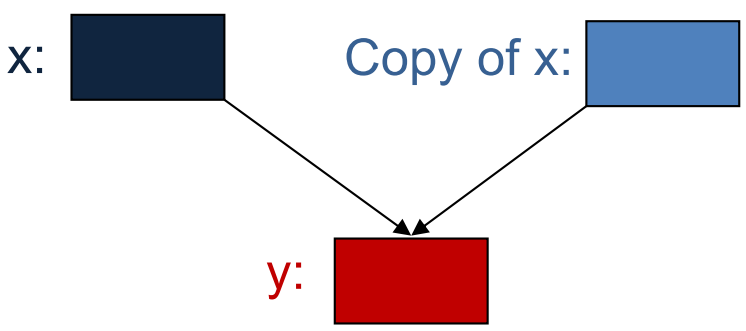
Copy Control and Resource Management

- **Like-a-value** classes have their own state
  - When we copy a like-a-value object, the copy and the original are independent of each other
  - Changes made to the copy have no effect on the original, and vice versa

- **Like-a-pointer** classes that act like pointers share part of the state
  - When we copy objects of such classes, the copy and the original use the same underlying data
  - Changes made to the copy also change the original, and vice versa

Copy + like-a-pointer / like-a-value semantic

- **Shallow copy → Like-a-pointer**
  - Copy only a pointer so that the two pointers now refer to the same object
  - What pointers and references do
- **Deep copy → Like-a-value**
  - What the pointer points to so that the two pointers now each refer to a distinct object
  - What vector, string, etc. do
  - Requires copy constructors and copy assignments for container classes
  - Must copy "all the way down" if there are more levels in the object



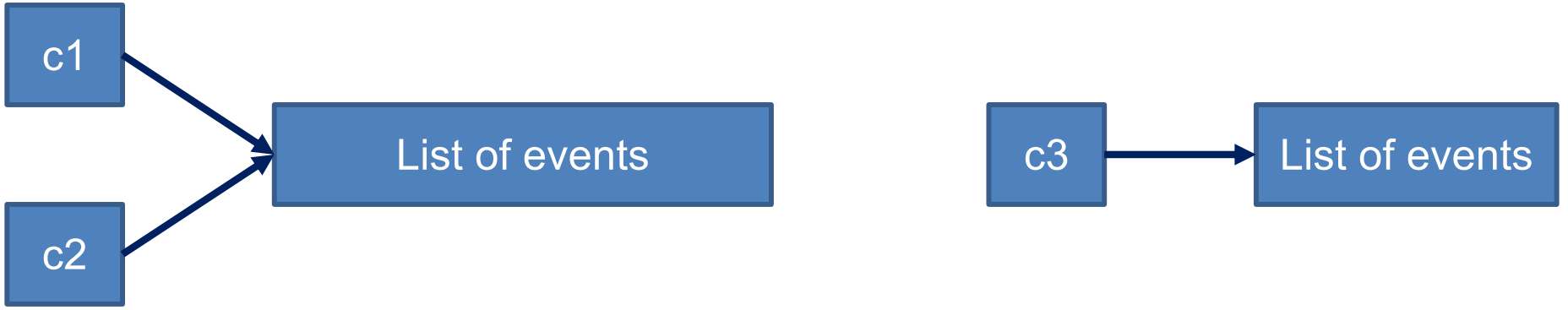
Shallow copy / like-a-pointer



Deep copy / like-a-value

Goal

- Implement a simple calendar
- A `Calendar` stores `Event`s; each `Event` is characterized by date (`time_t`) and name (`std::string`)
  - The class `Event` is given
- **Like-a-pointer** behavior



Required methods

- **`addEvent(const Event&)`**, that receives an `Event` and adds it to the calendar
- **`updateEvent(time_t, const string&, time_t, const string&)`**, that receives old date and name, finds the corresponding event and updates the information with the new values
- **`print()`**, that prints the name of all events in the calendar